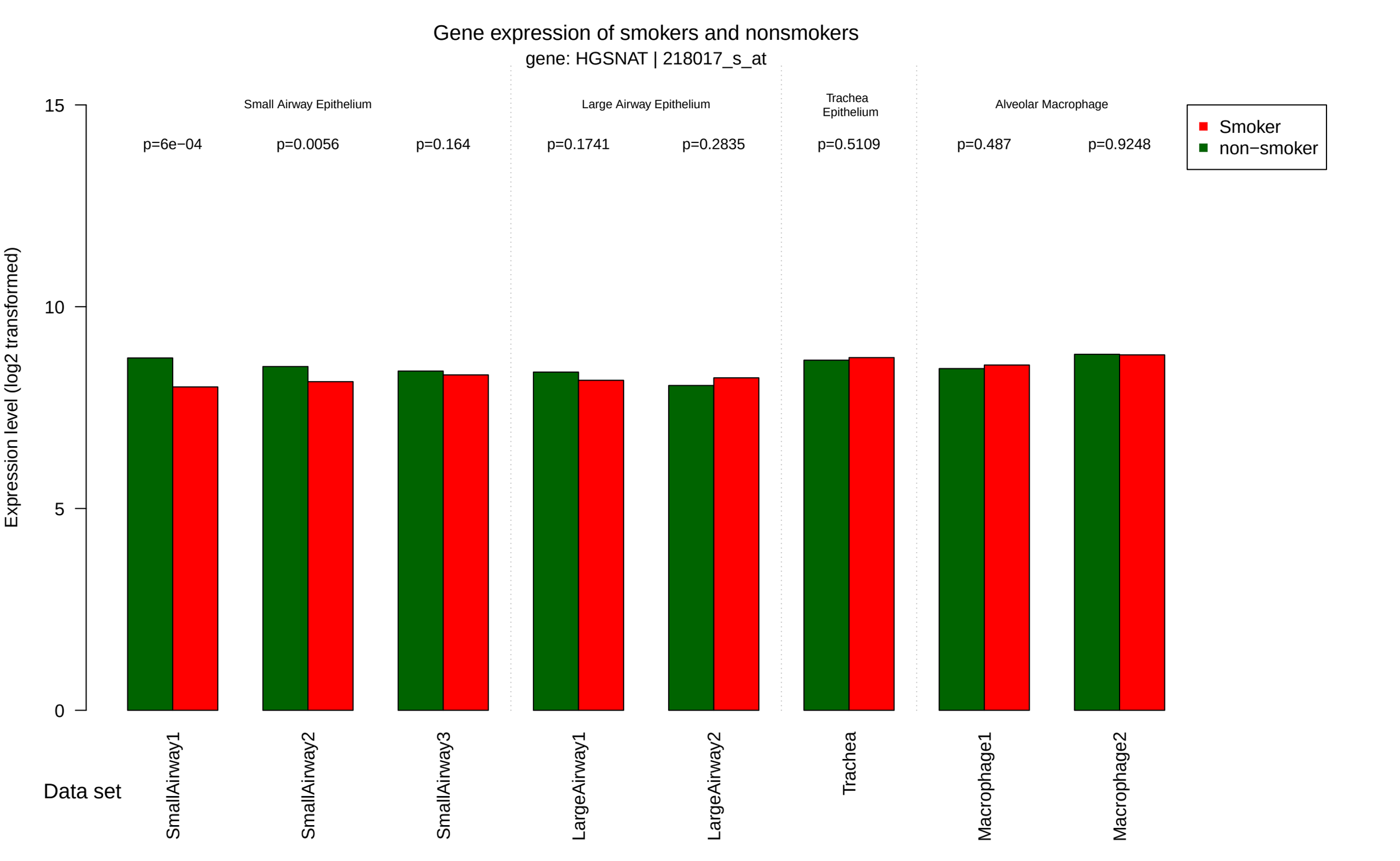
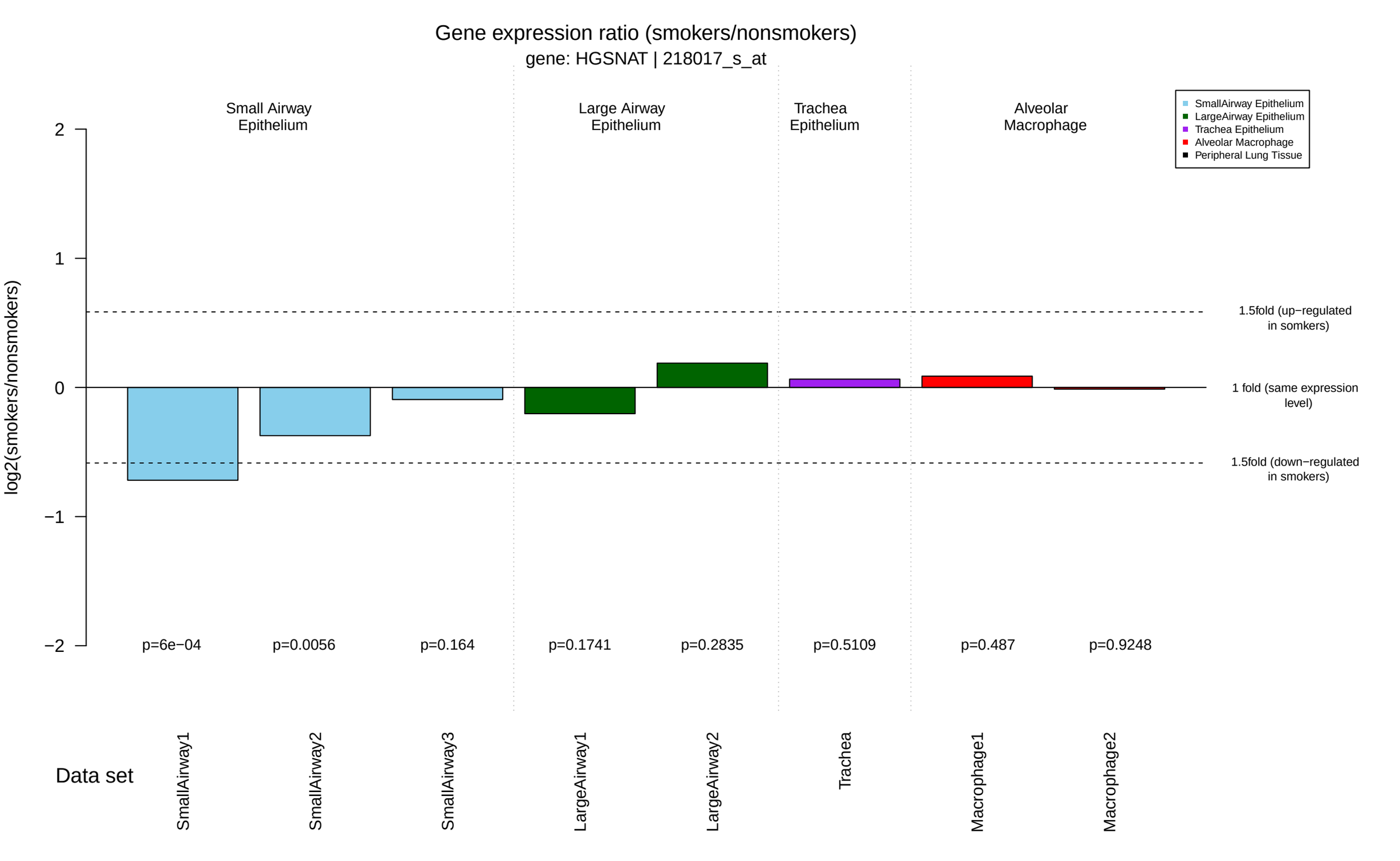
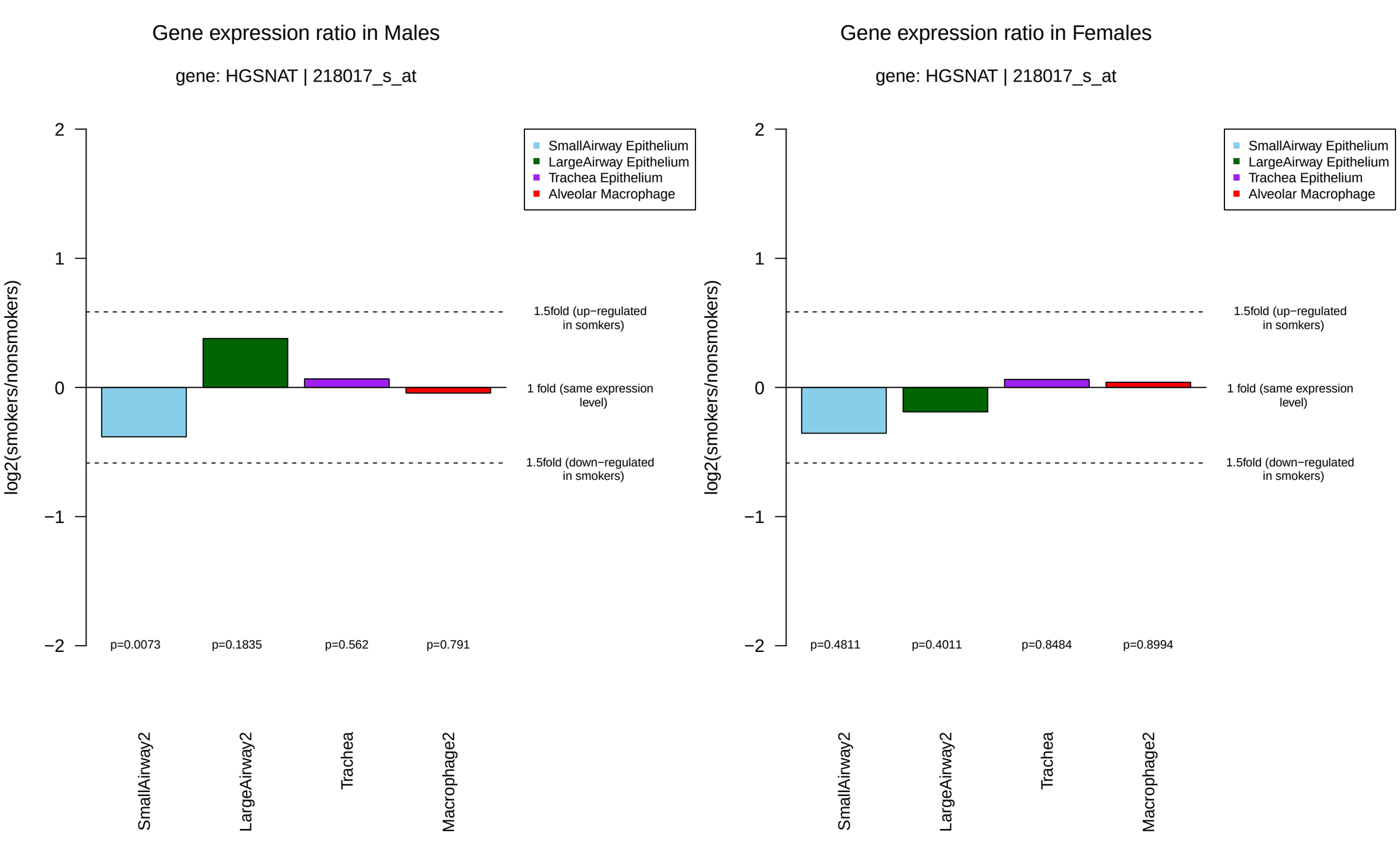
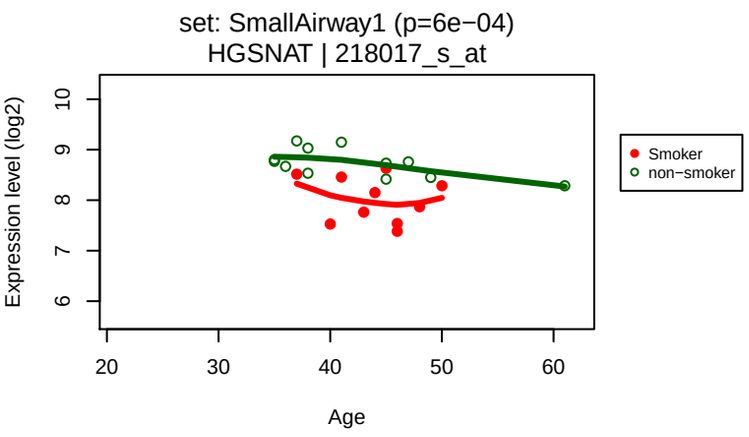
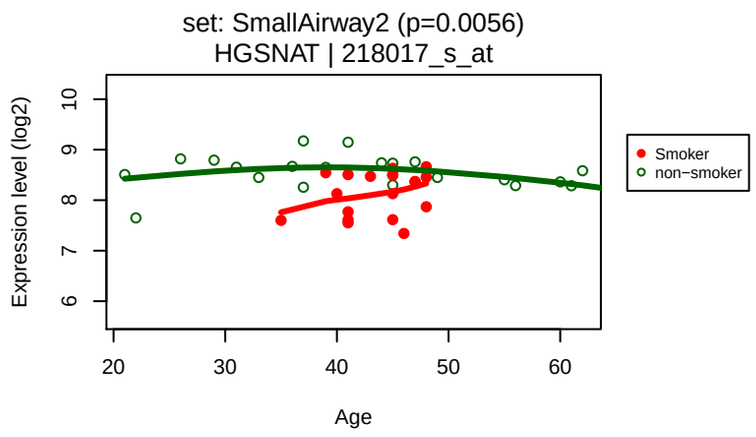
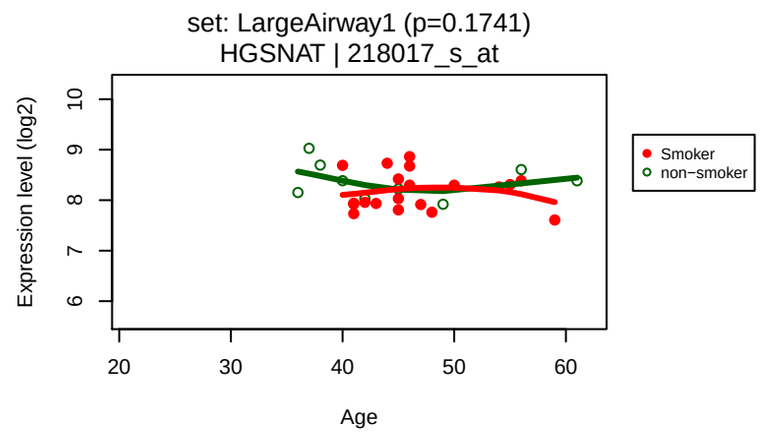

Gene expression of smokers and nonsmokers
gene: HGSNAT | 218017_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=6e−04
p=0.0056
p=0.164
p=0.1741
p=0.2835
p=0.5109
p=0.487
p=0.9248
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

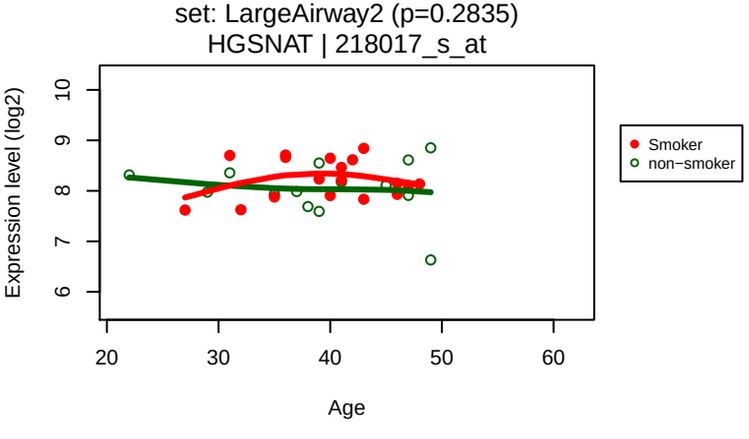

Gene expression ratio (smokers/nonsmokers)
gene: HGSNAT | 218017_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.0056
p=0.164
p=0.1741
p=0.2835
p=0.5109
p=0.487
p=0.9248
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

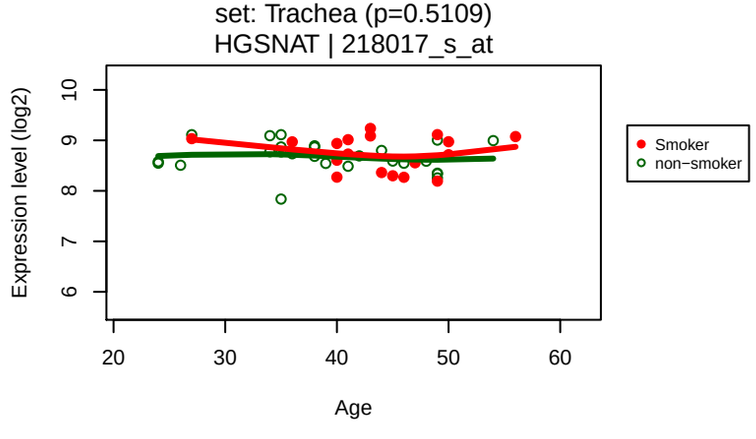

Gene expression ratio in Males
gene: HGSNAT | 218017_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0073
p=0.1835
p=0.562
p=0.791
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HGSNAT | 218017_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4811
p=0.4011
p=0.8484
p=0.8994
SmallAirway2
LargeAirway2
Trachea
Macrophage2

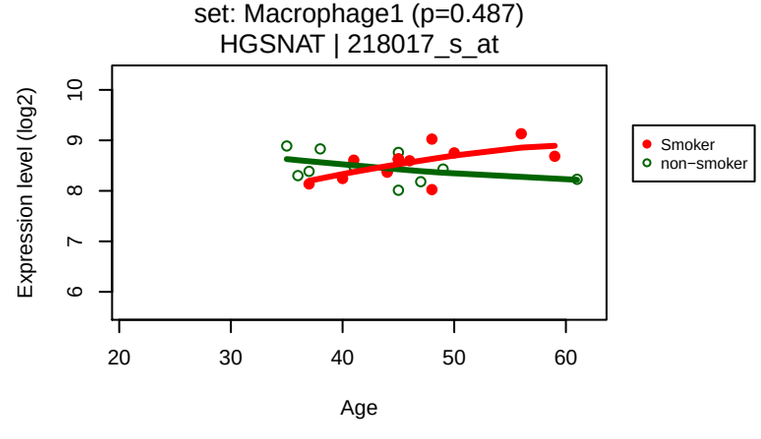

set: SmallAirway1 (p=6e−04)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker

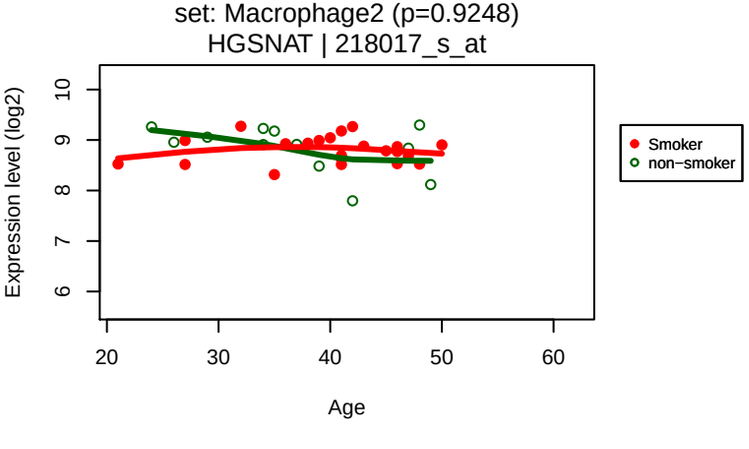

set: SmallAirway2 (p=0.0056)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker

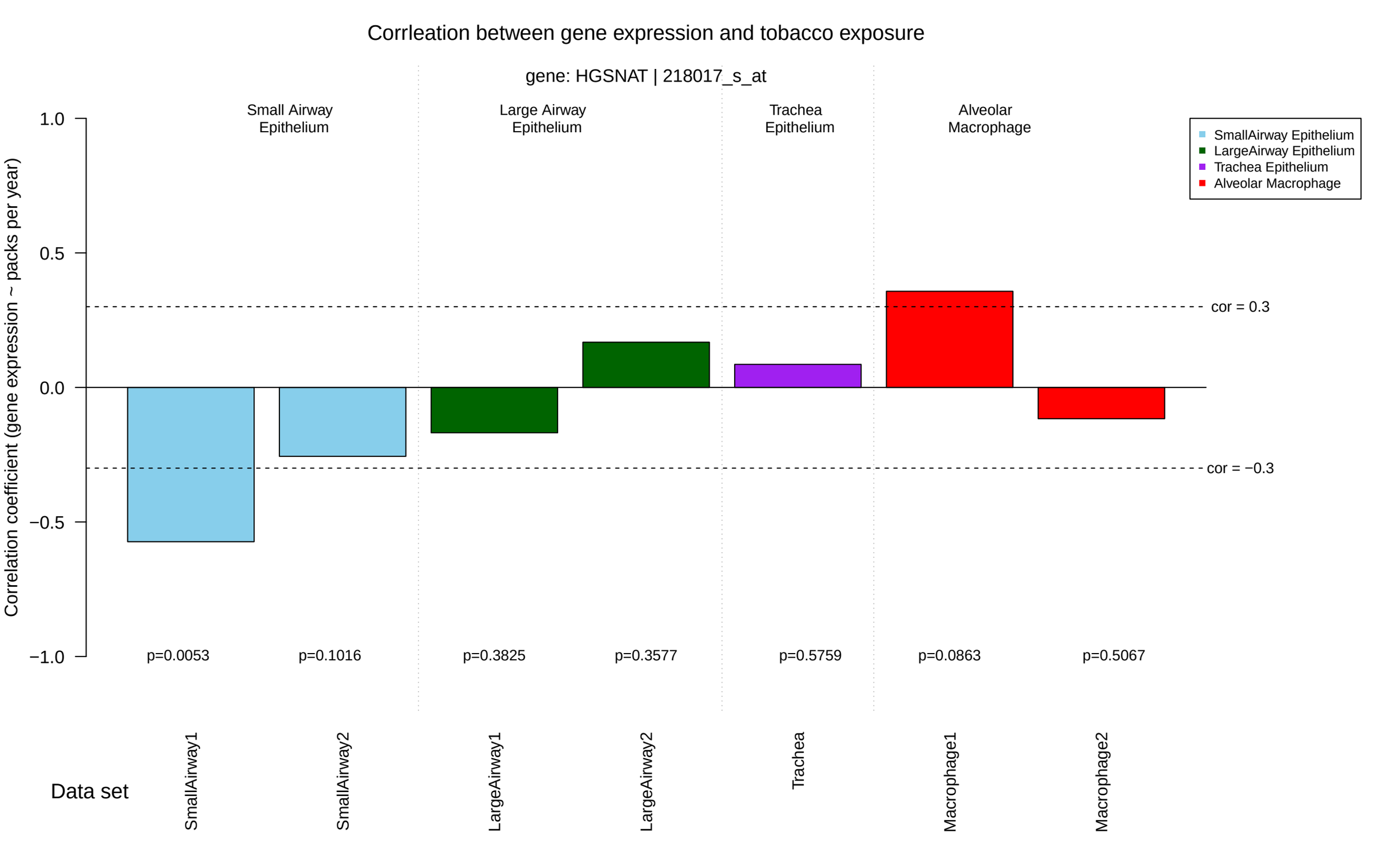

set: LargeAirway1 (p=0.1741)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2835)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5109)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.487)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9248)
HGSNAT | 218017_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HGSNAT | 218017_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0053
p=0.1016
p=0.3825
p=0.3577
p=0.5759
p=0.0863
p=0.5067
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2